

# P1149

## Diagnostic information - DME (DFI) control unit

Oil pressure control valve: power supply

### Diagnostic conditions

- Engine running

### Possible fault causes

- ♦ Open circuit/short circuit to ground (power supply to oil pressure control valve)
- ♦ Short circuit to B+/ground or open circuit in the line (oil pressure control valve activation)
- ♦ Oil pressure control valve faulty

## Diagnostic information - DME (DFI) control unit

### Oil pressure control valve: activation

#### **INFORMATION**

- ♦ The oil pressure control valve is controlled via a PWM signal.
- ♦ The PWM signal depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine (rpm, load, oil temperature), measured oil pressure (at the oil pressure sensor) ...
- ♦ A fault is detected if the PWM signal is outside a defined range for > 8 seconds.

#### **Diagnostic conditions**

- Engine running

#### **Possible fault causes**

- ♦ Short circuit to B+ (oil pressure control valve activation)
- ♦ Open circuit/short circuit to B+ (power supply to oil pressure control valve)
- ♦ Oil pressure control valve faulty

# P1151

## Diagnostic information - DME (DFI) control unit

### Oil pressure control valve: activation

#### **INFORMATION**

- ♦ The oil pressure control valve is controlled via a PWM signal.
- ♦ The PWM signal depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine (rpm, load, oil temperature), measured oil pressure (at the oil pressure sensor) ...
- ♦ A fault is detected if the PWM signal is outside a defined range for > 8 seconds.
- ♦ The oil pressure is reduced to the minimum pressure in the event of a short circuit to ground.

#### **Diagnostic conditions**

- Engine running

#### **Possible fault causes**

- ♦ Short circuit to ground (oil pressure control valve activation)
- ♦ Open circuit/short circuit to ground (power supply to oil pressure control valve)
- ♦ Oil pressure control valve faulty

#### **Fault effects**

- ♦ Power supply to oil pressure control valve - is switched off

## Diagnostic information - DME (DFI) control unit

### Oil pressure control valve: activation

#### **INFORMATION**

- ♦ The oil pressure control valve is controlled via a PWM signal.
- ♦ The PWM signal depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine (rpm, load, oil temperature), measured oil pressure (at the oil pressure sensor) ...
- ♦ A fault is detected if the PWM signal is outside a defined range for > 8 seconds.

#### **Diagnostic conditions**

- Engine running

#### **Possible fault causes**

- ♦ Open circuit (oil pressure control valve activation)
- ♦ Open circuit/short circuit to ground (power supply to oil pressure control valve)
- ♦ Oil pressure control valve faulty

# P1157

## **Diagnostic information - DME (DFI) control unit**

### **Engine compartment temperature sensor**

#### **Diagnostic conditions**

- Ignition on

#### **Possible fault causes**

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine compartment temperature sensor faulty

# P1158

## **Diagnostic information - DME (DFI) control unit**

### **Engine compartment temperature sensor**

#### **Diagnostic conditions**

- Ignition on

#### **Possible fault causes**

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine compartment temperature sensor faulty

# P1165

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 1: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P1166

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 1: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P1167

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 1: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P1168

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 2: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P1169

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 2: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P1170

## Diagnostic information - DME (DFI) control unit

### Boost pressure adjuster, bank 2: electrical fault

(Electronic adjuster for turbocharger)

#### Diagnostic conditions

- Ignition on, then:
  - Switch off ignition
  - Switch on ignition

#### Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Power supply or ground supply faulty
- ♦ Boost pressure adjuster (electronic adjuster for turbocharger) faulty

#### Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Reduced engine torque

# P117A

## Diagnostic information - DME (DFI) control unit

### Charge air temperature too high - cooling system fault

- Boxster and Cayman only

### Diagnostic conditions

- Engine running

### Possible fault causes

- ♦ Air intake for charge-air cooler covered (e.g. leaves, ...)
- ♦ Charge-air cooler damaged
- ♦ Air outlet from charge-air cooler covered
- ♦ Engine compartment purge fan has insufficient effect
- ♦ Electric coolant pumps have insufficient effect
- ♦ Air in the indirect charge air cooling system's low-temperature cooling circuit

### Fault setting condition(s)

- Intake air temperature above 90 °C (194 °F)

### Fault effects

- ♦ Warning in instrument cluster >Cooling system fault<
- ♦ Reduced engine torque

# P117B

## Diagnostic information - DME (DFI) control unit

### Charge air cooling - inadequate effect

- Carrera only

#### Diagnostic conditions

- Driving at a speed of over 125 km/h (77 mph) for more than 5 minutes with the engine compartment purge fan active
- No engine compartment purge fan or rear spoiler faults

#### Possible fault causes

- ♦ Air intake for charge-air cooler covered (e.g. leaves, rear spoiler does not extend, ...)
- ♦ Charge-air cooler damaged
- ♦ Air outlet from charge-air cooler covered

#### Fault setting condition(s)

- Intake air temperature is more than 50 °C (122 °C) higher than the ambient temperature

### Charge air cooling - inadequate effect

- Boxster and Cayman only

#### Diagnostic conditions

- Driving at a speed of over 30 km/h (18 mph) for more than 17 minutes or
- Driving at a speed of over 60 km/h (37 mph) for more than 4 minutes
- With charge air cooling active (engine compartment purge fan and electric coolant pumps active)
- No engine compartment purge fan or electric coolant pump faults

#### Possible fault causes

- ♦ Air intake for charge-air cooler covered (e.g. leaves, ...)
- ♦ Charge-air cooler damaged
- ♦ Air outlet from charge-air cooler covered
- ♦ Engine compartment purge fan has insufficient effect
- ♦ Electric coolant pumps have insufficient effect
- ♦ Air in the indirect charge air cooling system's low-temperature cooling circuit

#### Fault setting condition(s)

- The intake air temperature is significantly higher than the ambient temperature (more than 60 to 85 °C (140 to 185 °F), depending on the ambient temperature < 10 to > 40 °C (< 50 to > 104 °F)).

# P1186

## Diagnostic information - DME (DFI) control unit

### Ambient pressure sensor

#### Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible

#### Possible fault causes

- ♦ DME (DFI) control unit faulty

#### Fault effects

- ♦ Substitute value from model

# P1191

## Diagnostic information - DME (DFI) control unit

### LIN bus - radiator shutters

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Loose contact, short circuit or open circuit in LIN bus signal line between servo motors and DME control unit
- ♦ One servo motor faulty

#### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P1192

## Diagnostic information - DME (DFI) control unit

### LIN bus - radiator shutters

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Loose contact, short circuit or open circuit in LIN bus signal line between servo motors and DME control unit
- ♦ One servo motor faulty

#### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P1193

## Diagnostic information - DME (DFI) control unit

**Radiator shutter 1 (left): signal implausible or missing**

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Implausible request from DME control unit, e.g. during re-programming
- ♦ Loose contact, short circuit or open circuit in signal line between servo motor and DME control unit
- ♦ Servo motor power supply or ground supply faulty
- ♦ Servo motor faulty

### Fault setting conditions

- No signal or implausible signal on the signal line between servo motor and DME control unit

### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P1194

## Diagnostic information - DME (DFI) control unit

**Radiator shutter 2 (right): signal implausible or missing**

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Implausible request from DME control unit, e.g. during re-programming
- ♦ Loose contact, short circuit or open circuit in signal line between servo motor and DME control unit
- ♦ Servo motor power supply or ground supply faulty
- ♦ Servo motor faulty

### Fault setting conditions

- No signal or implausible signal on the signal line between servo motor and DME control unit

### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P1195

## Diagnostic information - DME (DFI) control unit

### Radiator shutter 1 (left): signal implausible or missing

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Implausible request from DME control unit, e.g. during re-programming
- ♦ Loose contact, short circuit or open circuit in signal line between servo motor and DME control unit
- ♦ Servo motor power supply or ground supply faulty
- ♦ Servo motor faulty

#### Fault setting conditions

- No signal or implausible signal on the signal line between servo motor and DME control unit

#### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P1196

## Diagnostic information - DME (DFI) control unit

**Radiator shutter 2 (right): signal implausible or missing**

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Implausible request from DME control unit, e.g. during re-programming
- ♦ Loose contact, short circuit or open circuit in signal line between servo motor and DME control unit
- ♦ Servo motor power supply or ground supply faulty
- ♦ Servo motor faulty

### Fault setting conditions

- No signal or implausible signal on the signal line between servo motor and DME control unit

### Fault effects

- ♦ If it still functions, the servo motor opens the shutter as far as possible

# P119A

## Diagnostic information - DME (DFI) control unit

**Intake manifold pressure sensor faulty**

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Intake manifold pressure sensor faulty

### Fault effects

- ♦ Substitute value from model

# P119E

## Diagnostic information - DME (DFI) control unit

**Intake manifold pressure sensor: electrical fault**

### Diagnostic conditions

- Ignition on or engine running

### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Intake manifold pressure sensor faulty

### Fault effects

- ♦ Substitute value from model

# P11A1

## **Diagnostic information - DME (DFI) control unit**

**Boost pressure sensor (pressure sensor ahead of throttle valve) faulty**

### **Diagnostic conditions**

- Ignition on

### **Possible fault causes**

- ♦ Pressure sensor ahead of throttle valve faulty

### **Fault effects**

- ♦ Substitute value from model

# P11A5

## Diagnostic information - DME (DFI) control unit

**Boost pressure sensor (pressure sensor ahead of throttle valve): electrical fault**

### Diagnostic conditions

- Ignition on or engine running

### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Pressure sensor ahead of throttle valve faulty

### Fault effects

- ♦ Substitute value from model

# P11A8

## **Diagnostic information - DME (DFI) control unit**

**Exhaust temperature sensor, bank 1 faulty**

### **Diagnostic conditions**

- Ignition on

### **Possible fault causes**

- ♦ Exhaust temperature sensor, bank 1 faulty

# P11A9

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 1 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 1 faulty

# P11AA

## **Diagnostic information - DME (DFI) control unit**

**Exhaust temperature sensor, bank 1 faulty**

### **Diagnostic conditions**

- Ignition on

### **Possible fault causes**

- ♦ Exhaust temperature sensor, bank 1 faulty

# P11AB

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 1 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 1 faulty

# P11AC

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 1 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 1 faulty

# P11AF

## **Diagnostic information - DME (DFI) control unit**

**Exhaust temperature sensor, bank 2 faulty**

### **Diagnostic conditions**

- Ignition on

### **Possible fault causes**

- ♦ Exhaust temperature sensor, bank 2 faulty

# P11B0

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 2 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 2 faulty

# P11B1

## **Diagnostic information - DME (DFI) control unit**

**Exhaust temperature sensor, bank 2 faulty**

### **Diagnostic conditions**

- Ignition on

### **Possible fault causes**

- ♦ Exhaust temperature sensor, bank 2 faulty

# P11B2

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 2 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 2 faulty

# P11B3

## Diagnostic information - DME (DFI) control unit

### Exhaust temperature sensor, bank 2 - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Exhaust temperature sensor, bank 2 faulty

# P11B6

## Diagnostic information - DME (DFI) control unit

**Oil pressure sensor faulty**

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Oil pressure sensor faulty

# P11B7

## Diagnostic information - DME (DFI) control unit

### Oil pressure sensor - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Oil pressure sensor faulty

# P11B8

## Diagnostic information - DME (DFI) control unit

Oil pressure sensor faulty

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Oil pressure sensor faulty

# P11B9

## Diagnostic information - DME (DFI) control unit

### Oil pressure sensor - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Oil pressure sensor faulty

# P11BA

## Diagnostic information - DME (DFI) control unit

### Oil pressure sensor - electrical fault

#### Diagnostic conditions

- Ignition on or engine running

#### Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ 5 V power supply faulty
- ♦ Ground supply faulty
- ♦ Oil pressure sensor faulty